

Static member in C++

DATE



- Static local Variable ÷ Is a variable declared within a function

```
#include <iostream>
```

```
using namespace std;
```

```
void fun()
```

```
{
```

```
    static int x; → By default initialize with zero
```

```
    int y; → contain garbage value
```

```
}
```

Important ÷

- y get memory when the function fun() is called.
But x get memory at the starting of the program

- y get destroyed as fun() work is completed.
But x remains throughout the end of the program.

* Static variable get initialize only once then after it retains its value

- Static member variable

A static member variable is a variable declared within a class using the static keyword.



Static Member Variable

```
#include <iostream>
```

```
using namespace std;
```

```
class Account
```

```
{
```

```
private:
```

```
int balance; → Instance Member variable
```

```
static float xoi; → static Member variable / Class variable
```

```
public:
```

```
void setbalance(int b)
```

```
{
```

```
balance = b;
```

```
}
```

```
};
```

```
float Account::xoi;
```

Syntax ÷

Variable type Membership label variable name

In C++ by just **declaring** variable as static it is not formed

We need to define **variable** outside the **Class**.

```
void main()
```

```
{
```

```
Account a1;
```

In object **a1** there will be only one variable **balance** is present.

Static member variable (xoi) is ~~the~~ not the part of object **a1**.

* There will be only one copy of **static member variable** for the whole class

Static member variable doesn't depend on **object**

Static Member ~~function~~ Variable :

As we can access static member ~~function~~ ^{variable} with the help of object and class name.

```
#include <iostream>
using namespace std;
class Account
{
    public:
        int balance;
        static float xoi;

        void setbalance(int b)
        {
            balance = b;
        }
};

float Account :: xoi;
void main()
{
    Account a1;
    a1.xoi = 4.5f; // Accessing static member with object
    Account :: xoi = 6.5f; // Accessing static member with Class
}
```

But if static member is private then we can't access it through object and class.

Then we have to make ~~it~~ function to access it.



```
#include <iostream>
```

```
using namespace std;
```

```
class Account
```

```
{
```

```
private:
```

```
int balance;
```

```
static float roi;
```

```
public:
```

```
void setBalance(int b)
```

```
{
```

```
balance = b;
```

```
}
```

```
void setRoi(float x) // Instance Member function
```

```
{
```

```
roi = x;
```

```
}
```

```
};
```

```
float Account::roi;
```

```
void main()
```

```
{
```

```
Account a1;
```

```
a1.setRoi(4.5f);
```

```
}
```

As we know **static member variable** don't depend on object.
So if there is no object still **static member variable** exist.

So in case of no object to access static member variable
we use **static member function**.

Non-static member can't be call by **Class name**.



Static Member function:

```
#include <iostream>
using namespace std;
class Account
{
```

* A Static Member function can have access to only other static member declared in the same class

```
private:
    int balance;
    static float xoi;
```

```
public:
    void setBalance(int b)
    {
```

```
        balance = b;
```

```
    }
```

```
    static void setXoi(float x) // Static Member function
    {
```

```
        xoi = x;
```

```
    }
```

```
};
```

```
float Account::xoi;
```

```
void main()
```

```
{
```

```
    Account::setXoi(45f);
```

```
}
```